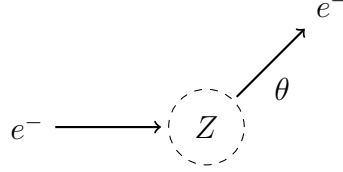


# Mott scattering

Mott scattering is the result of kinetic electrons  $e^-$  interacting with atomic nuclei  $Z$ .



Momentum vectors where  $p$  is electron momentum and  $E$  is relativistic energy  $E = \sqrt{p^2 + m^2}$ .

$$p_1 = (E, \mathbf{p}_1) = \begin{pmatrix} E \\ 0 \\ 0 \\ p \end{pmatrix} \quad \text{inbound electron} \quad p_2 = (E, \mathbf{p}_2) = \begin{pmatrix} E \\ p \sin \theta \cos \phi \\ p \sin \theta \sin \phi \\ p \cos \theta \end{pmatrix} \quad \text{outbound electron}$$

Spinors for the inbound electron. All spinors are normalized as  $u^\dagger u = 2E$ .

$$u_{11} = \frac{1}{\sqrt{E+m}} \begin{pmatrix} E+m \\ 0 \\ p \\ 0 \end{pmatrix} \quad \text{inbound electron spin up} \quad u_{12} = \frac{1}{\sqrt{E+m}} \begin{pmatrix} 0 \\ E+m \\ 0 \\ -p \end{pmatrix} \quad \text{inbound electron spin down}$$

Spinors for the outbound electron.

$$u_{21} = \frac{1}{\sqrt{E+m}} \begin{pmatrix} E+m \\ 0 \\ p_{2z} \\ p_{2x} + ip_{2y} \end{pmatrix} \quad \text{outbound electron spin up} \quad u_{22} = \frac{1}{\sqrt{E+m}} \begin{pmatrix} 0 \\ E+m \\ p_{2x} - ip_{2y} \\ -p_{2z} \end{pmatrix} \quad \text{outbound electron spin down}$$

Inbound wave functions. All wave functions are normalized as  $\langle \psi | \psi \rangle = 1$  over volume  $V$ .

$$\psi_{11}(\mathbf{r}) = \frac{1}{\sqrt{2EV}} u_{11} \exp\left(\frac{i}{\hbar} \mathbf{p}_1 \cdot \mathbf{r}\right) \\ \psi_{12}(\mathbf{r}) = \frac{1}{\sqrt{2EV}} u_{12} \exp\left(\frac{i}{\hbar} \mathbf{p}_1 \cdot \mathbf{r}\right)$$

Outbound wave functions.

$$\psi_{21}(\mathbf{r}) = \frac{1}{\sqrt{2EV}} u_{21} \exp\left(\frac{i}{\hbar} \mathbf{p}_2 \cdot \mathbf{r}\right) \\ \psi_{22}(\mathbf{r}) = \frac{1}{\sqrt{2EV}} u_{22} \exp\left(\frac{i}{\hbar} \mathbf{p}_2 \cdot \mathbf{r}\right)$$

For the Coulomb potential

$$V(\mathbf{r}) = -\frac{Ze^2}{4\pi\epsilon_0 r} = -\frac{Z\alpha}{r}$$

we have the following transition amplitudes.

$$\begin{aligned} M_1 &= -Z\alpha \int \frac{1}{r} \psi_{21}^\dagger(\mathbf{r}) \psi_{11}(\mathbf{r}) d^3\mathbf{r} && \text{up} \leftarrow \text{up} \\ M_2 &= -Z\alpha \int \frac{1}{r} \psi_{21}^\dagger(\mathbf{r}) \psi_{12}(\mathbf{r}) d^3\mathbf{r} && \text{up} \leftarrow \text{down} \\ M_3 &= -Z\alpha \int \frac{1}{r} \psi_{22}^\dagger(\mathbf{r}) \psi_{11}(\mathbf{r}) d^3\mathbf{r} && \text{down} \leftarrow \text{up} \\ M_4 &= -Z\alpha \int \frac{1}{r} \psi_{22}^\dagger(\mathbf{r}) \psi_{12}(\mathbf{r}) d^3\mathbf{r} && \text{down} \leftarrow \text{down} \end{aligned}$$

Substitute for the wave functions.

$$\begin{aligned} M_1 &= -\frac{1}{2EV} u_{21}^\dagger u_{11} Z\alpha \int \frac{1}{r} \exp \left[ \frac{i}{\hbar} (\mathbf{p}_1 - \mathbf{p}_2) \cdot \mathbf{r} \right] d^3\mathbf{r} \\ M_2 &= -\frac{1}{2EV} u_{21}^\dagger u_{12} Z\alpha \int \frac{1}{r} \exp \left[ \frac{i}{\hbar} (\mathbf{p}_1 - \mathbf{p}_2) \cdot \mathbf{r} \right] d^3\mathbf{r} \\ M_3 &= -\frac{1}{2EV} u_{22}^\dagger u_{11} Z\alpha \int \frac{1}{r} \exp \left[ \frac{i}{\hbar} (\mathbf{p}_1 - \mathbf{p}_2) \cdot \mathbf{r} \right] d^3\mathbf{r} \\ M_4 &= -\frac{1}{2EV} u_{22}^\dagger u_{12} Z\alpha \int \frac{1}{r} \exp \left[ \frac{i}{\hbar} (\mathbf{p}_1 - \mathbf{p}_2) \cdot \mathbf{r} \right] d^3\mathbf{r} \end{aligned}$$

Solve the integral to obtain

$$\int \frac{1}{r} \exp \left[ \frac{i}{\hbar} (\mathbf{p}_1 - \mathbf{p}_2) \cdot \mathbf{r} \right] d^3\mathbf{r} = \frac{\pi \hbar^2}{p^2 \sin^2(\theta/2)} \quad (1)$$

Hence

$$\begin{aligned} M_1 &= -\frac{1}{2EV} u_{21}^\dagger u_{11} \frac{Z\alpha\pi\hbar^2}{p^2 \sin^2(\theta/2)} \\ M_2 &= -\frac{1}{2EV} u_{21}^\dagger u_{12} \frac{Z\alpha\pi\hbar^2}{p^2 \sin^2(\theta/2)} \\ M_3 &= -\frac{1}{2EV} u_{22}^\dagger u_{11} \frac{Z\alpha\pi\hbar^2}{p^2 \sin^2(\theta/2)} \\ M_4 &= -\frac{1}{2EV} u_{22}^\dagger u_{12} \frac{Z\alpha\pi\hbar^2}{p^2 \sin^2(\theta/2)} \end{aligned}$$

Expected density  $\langle |M|^2 \rangle$  is the average density. (There are two inbound states hence the divisor is 2.)

$$\langle |M|^2 \rangle = \frac{1}{2} (|M_1|^2 + |M_2|^2 + |M_3|^2 + |M_4|^2)$$

Pull common factor from densities.

$$\langle |M|^2 \rangle = \left( \frac{1}{2EV} \frac{Z\alpha\pi\hbar^2}{p^2 \sin^2(\theta/2)} \right)^2 \frac{1}{2} \left( |u_{21}^\dagger u_{11}|^2 + |u_{21}^\dagger u_{12}|^2 + |u_{22}^\dagger u_{11}|^2 + |u_{22}^\dagger u_{12}|^2 \right)$$

By computer algebra

$$\frac{1}{2} \left( |u_{21}^\dagger u_{11}|^2 + |u_{21}^\dagger u_{12}|^2 + |u_{22}^\dagger u_{11}|^2 + |u_{22}^\dagger u_{12}|^2 \right) = 4m^2 + 4p^2 \cos^2(\theta/2) \quad (2)$$

Hence

$$\langle |M|^2 \rangle = \left( \frac{1}{2EV} \frac{Z\alpha\pi\hbar^2}{p^2 \sin^2(\theta/2)} \right)^2 (4m^2 + 4p^2 \cos^2(\theta/2))$$

By Fermi's golden rule

$$\frac{d\sigma}{d\Omega} = \left( \frac{EV}{2\pi\hbar^2} \right)^2 \langle |M|^2 \rangle$$

Hence

$$\frac{d\sigma}{d\Omega} = \left( \frac{Z\alpha}{4p^2 \sin^2(\theta/2)} \right)^2 (4m^2 + 4p^2 \cos^2(\theta/2)) \quad (3)$$

Factor the  $4m^2$  to obtain

$$\frac{d\sigma}{d\Omega} = \left( \frac{Z\alpha m}{2p^2 \sin^2(\theta/2)} \right)^2 (1 + v^2 \cos^2(\theta/2)) \quad (4)$$

where  $v = p/m$ . Noting that

$$\frac{d\sigma}{d\Omega}_{\text{Rutherford}} = \left( \frac{Z\alpha m}{2p^2 \sin^2(\theta/2)} \right)^2$$

we have

$$\frac{d\sigma}{d\Omega} = \frac{d\sigma}{d\Omega}_{\text{Rutherford}} (1 + v^2 \cos^2(\theta/2))$$

Eigenmath script